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Drug injection control device for temporarily maintaining twist of torsion spring to which rotational force is applied through dose dial

Abstract

A drug injection control device includes a housing, a dose dial, a cartridge holder fixedly coupled to an opposite side of the housing, an injection button positioned in the dose dial, an injection amount setting module, and a drug delivery module, wherein the injection amount setting module includes a reset tube switchably coupled to the dose dial by the operation of the injection button, a ratchet tube coupled to an outer side of the reset tube to rotate at the same time, a scale drum which slidably rotates along an axial direction on an outer surface of the ratchet tube, a torsion spring having two ends coupled to the housing and the ratchet tube, and an anchor rotatably coupled to the ratchet tube and slidably coupled to an inside of the housing.

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Background/Summary

PRIORITY

(1) This application claims benefit under 35 U.S.C. 119, 120, 121, or 365(c), and is a National Stage entry from International Application No. PCT/KR2021/015683, filed Nov. 2, 2021, which claims priority to the benefit of Korean Patent Application No. 10-2020-0145239 filed in the Korean Intellectual Property Office on Nov. 3, 2020, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a drug injection control device for temporarily maintaining the twist of a torsion spring to which a rotational force is applied through a dose dial, through an anchor configured to be rotatably coupled to a tube which moves with the dose dial while being selectively connected to a housing.

2. Background Art

(3) Autoinjectors are devices for easy self-administration by patients or even untrained personnel, and shield the patients from being exposed to syringe needles. There is a need for autoinjectors that are easy to use, acceptably small in size and simple to improve the manufacturing, assembly and use, resulting in reduced manufacturing and assembly costs, and have the minimized influence on the environment.

(4) Drug injectors used for subcutaneous injection require highly specialized skills and experience to use them, and recently, automatic drug injectors have been proposed, in which drugs are automatically injected via syringe needles that penetrate into the skin when pressed against the skin.

(5) Among the commercially available syringes for drug injection, pen-type syringes have functionality to feed multiple doses from drug containers, and the dose quantity is set immediately before the dose delivery. Since most of drug delivery device users have little experience, in some instances, it is desirable to provide the users with some types of indicators or information notifying that the dose delivery sequences were carried out and it is safe to remove the drug delivery devices from the dose delivery sites.

(6) Korean Patent Publication No. 10-2017-00065623 discloses a drug injector device including a housing having an internal space and a lengthwise window, a rotatable dose dial that is maintained in the axial direction with respect to the housing, and a rotatable scale drum to present an indication for displaying the size of the set dose, wherein a sliding element of the scale drum is configured to slide in the axial direction with respect to the housing during the dose setting, the indication presented by the scale drum can be seen through a sliding window and accordingly the lengthwise window and the sliding window are combined with the indication to form a dose size display, the rotatable scale drum is rotated in the internal space formed by the housing during the dose setting, and an inner surface of the sliding element has inner features coupled to outer screw threads on an outer surface of the scale drum.

(7) Korean Patent Publication No. 10-2019-0017806 discloses a drug injection device, and more particularly, a drug delivery device including a housing and an actuation mechanism arranged inside the housing, wherein the actuation mechanism includes a plunger rod arranged to act on a stopper of a drug container and an actuator arranged slidable and connected to the plunger rod for acting on the stopper when the actuator is operated by displacing it in a longitudinal direction of the actuation mechanism.

SUMMARY

(8) The present disclosure is designed to solve the above-described problem of the conventional technology, and therefore the present disclosure is directed to providing a drug injection control device technology for temporarily maintaining the twist of a torsion spring to which a rotational force is applied through a dose dial, through an anchor configured to be rotatably coupled to a tube

which moves with the dose dial while being selectively connected to a housing when setting an injection amount of drug through the dose dial and injecting the set drug through an injection button.

(9) To achieve the above-described object, a drug injection control device according to the present disclosure includes a hollow housing **10**, a dose dial **20** rotatably coupled to a side of the housing **10**, a cartridge holder **30** fixedly coupled to an opposite side of the housing **10** wherein a cartridge containing a drug is received in the cartridge holder **30** and an injection needle is coupled to a side to deliver the drug, and an injection button **40** positioned in the dose dial **20**, and includes an injection amount setting module **100** to set an amount of the drug to be injected by storing a twisting moment in a torsion spring **140** positioned in the housing **10**; and a drug delivery module **200** to deliver the drug from the cartridge embedded in the cartridge holder **30** by moving through the release of the twisting moment applied to the torsion spring **140**, and preferably, the injection amount setting module **100** may include a reset tube **110** switchably coupled to the dose dial **20** by the operation of the injection button **40**, a ratchet tube **120** coupled to an outer side of the reset tube **110** to rotate at the same time, a scale drum **130** which slidably rotates along an axial direction on an outer surface of the ratchet tube **120**, a torsion spring **140** having two ends coupled to the housing **10** and the ratchet tube **120**, and an anchor **170** rotatably coupled to the ratchet tube **120** and slidably coupled to an inside of the housing **10**, the anchor **170** may include an anchor circumferential protrusion **171** protruding along a circumferential direction on an inner surface, a tube rotation prevention protrusion **173** stepped in a teeth shape along the axial direction on the inner surface, and housing coupling protrusions **172**, **174** formed along the axial direction on the outer circumferential surface, and the tube rotation prevention protrusion **173** may be coupled to a rotation prevention hole **124** which constitutes the ratchet tube **120** to allow the ratchet tube **120** rotatably coupled to the anchor **170** to rotate in one direction only.

(10) Preferably, the anchor **170** may be rotatably coupled to an anchor coupling groove **122** formed on the ratchet tube **120** through the anchor circumferential protrusion **171**, and decouplably coupled to an anchor sliding portion **13** formed in the housing **10** through the housing coupling protrusions

(11) Preferably, the rod guide **210** coupled to the reset tube **110** may be connected to an inside of the piston rod **220** by screw rotation coupling to allow a sub-linear motion of the piston rod **220** through the rotation of the rod guide **210**, thereby having a cartridge drug release structure through the screw coupling between the piston rod **220** and the rod guide **210**.

(12) Preferably, the torsion spring **140** may have two ends coupled to the spring base **150** immovably coupled to the housing **10** and the ratchet tube **120**, respectively, and during the rotation through the dose dial **20**, a side of the torsion spring **140** fixed to the ratchet tube **120** may rotate, while an opposite side of the torsion spring **140** fixed to the spring base **150** may be kept in fixed state, to get the spring twisted.

(13) Preferably, the button plate **160** may be fixed to a plate fixing step **24** formed along the circumferential direction on an inner surface of the dose dial **20** on a first dial chamber formed through a partition wall in the dose dial **20**. Preferably, the injection amount setting module **100** may include a button plate **160** fixed and positioned in the dose dial **20** to perform a function of supporting the button spring **50**, the injection button **40** may be slidably coupled through the button plate **160**, the drug delivery module **200** may include a rod guide **210** switchably coupled to the injection amount setting module **100** by the operation of the injection button **40**, and a piston rod **220** screw-coupled to an outer side of the rod guide **210** to deliver the drug from the cartridge embedded in the cartridge holder **30** by making a linear motion with a rotational motion of the rod guide **210**, and the injection button **40** may rotate simultaneously with the rotation of the dose dial **20**, but may allow a stable pushing motion through the button plate **160**.

(14) The drug injection control device according to the present disclosure as described above temporarily maintains the twist of the torsion spring to which a rotational force is applied through the dose dial, through the anchor configured to be rotatably coupled to the tube which moves with

the dose dial while being selectively connected to the housing, when the injection amount of drug is preset according to the user's intention through the injection amount setting module configured to make a rotational motion with the dose dial in the housing to set the amount of drug to be injected by storing the twisting moment in the torsion spring.

(15) Additionally, the rod guide is connected to the inside of the piston rod by screw rotation coupling through the reset tube movably coupled to the injection button to allow a sub-linear motion of the piston rod through the rotation of the rod guide, thereby having a stable cartridge drug release structure through screw coupling between the piston rod and the rod guide.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an assembled perspective view of a drug injection control device according to an embodiment of the present disclosure.
- (2) FIG. 2 is an exploded perspective view of the drug injection control device according to FIG. 1.
- (3) FIG. 3 is a cross-sectional view of the drug injection control device of FIG. 1.
- (4) FIG. 4 is an enlarged view of the central part of FIG. 3.
- (5) FIG. 5 shows the internal part of a housing.
- (6) FIG. 6 shows an anchor rotatably coupled to a ratchet tube.
- (7) FIG. 7 shows the respective shape of a ratchet tube.
- (8) FIG. 8 shows the respective shape of an anchor.
- (9) FIG. 9 is a cross-sectional view showing a connection structure of a ratchet tube and an anchor.
- (10) FIG. 10 is a diagram showing a connection relationship between a dose dial and a reset tube.
- (11) FIG. 11 shows an injection button, a button plate, a reset tube, a spring base and a scale drum assembled together.
- (12) FIG. 12 is a perspective view of a spring base.
- (13) FIG. 13 shows a dose dial, a scale drum, a ratchet tube, a reset tube and a piston rod assembled together.
- (14) FIG. 14 shows an injection button, a button plate and a dose dial assembled together.
- (15) FIG. 15 is an enlarged view of the left part of FIG. 3.
- (16) FIG. 16 shows a dose dial.

DETAILED DESCRIPTION

(17) Hereinafter, the embodiments of the present disclosure will be described in more detail with reference to the accompanying drawings. However, the present disclosure is not limited to the following disclosed embodiments and will be embodied in a variety of different forms, and these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the present disclosure to those skilled in the art. In the drawings, like reference signs denote like elements.

(18) In adding the reference signs to the elements in each drawing, it should be noted that like elements have like reference signs as possible although they are shown in different drawings. Additionally, in describing the present disclosure, when it is determined that a certain detailed description of relevant known elements or functions may obscure the subject matter of the present disclosure, its detailed description is omitted.

(19) Hereinafter, a drug injection control device having a structure in which a rod guide **210** rotates and a piston rod moving with the rod guide **210** makes a linear motion to stably inject a drug set through a dose dial according to an embodiment of the present disclosure will be described with reference to FIGS. **1** to **16**.

(20) The drug injection control device includes a hollow housing **10**, a dose dial **20** rotatably coupled to one side of the housing **10**, a cartridge holder **30** fixedly coupled to the other side of the

housing **10**, wherein a cartridge containing a drug is received in the cartridge holder **30** and an injection needle is coupled to one side to deliver the drug, an injection button **40** positioned on the outer side of the dose dial **20**, a button spring **50** to allow the injection button **40** to be pressed and restored in the dose dial **20**, and a cap **60** separably coupled to the outside of the cartridge holder **30**.

(21) The present disclosure includes an injection amount setting module **100** configured to make a rotational motion with the dose dial **20** in the housing **10** to set an amount of drug to be injected by storing the twisting moment in a torsion spring; and a drug delivery module **200** switchably coupled to the injection amount setting module **100** by the operation of the injection button **40** to deliver the amount of drug set by the injection amount setting module **100** from the cartridge embedded in the cartridge holder **30** by moving through the release of the twisting moment applied to the torsion spring.

(22) The injection amount setting module **100** includes a reset tube **110** switchably coupled to the dose dial **20** by the operation of the injection button **40**, a ratchet tube **120** coupled to the outer side of the reset tube **110** to rotate at the same time, a scale drum **130** which slidably rotates along the axial direction on the outer surface of the ratchet tube **120**, a torsion spring **140** having two ends coupled to the housing **10** and the ratchet tube **120**, a spring base **150** to which the other side of the torsion spring **140** opposite to one side coupled to the ratchet tube **120** is fixed, a button plate **160** fixed and positioned in the dose dial **20** to perform the function of supporting the button spring **50**, and an anchor **170** rotatably coupled to the ratchet tube **120** in one direction and slidably coupled to the inside of the housing **10** along the axial direction.

(23) The reset tube **110** is a structure having a hollow cylindrical shape, and includes a button coupling groove **111** formed along the circumferential direction on the outer circumferential surface for coupling of button legs **41** which constitute the injection button **40**, a dial coupling protrusion **112** decouplably coupled to a tube coupling protrusion formed inside of the dose dial **20**, and a first tube coupling hole **113** coupled to the ratchet tube **120** to allow the ratchet tube **120** to rotate at the same time.

(24) The ratchet tube **120** is a structure having a hollow cylindrical shape for receiving the reset tube **110** therein, and includes a drum guide protrusion **121** protruding along the lengthwise direction on the outer circumferential surface, an anchor coupling groove **122** formed along the circumferential direction to allow rotatable coupling of the anchor **170** on one side of the ratchet tube **120**, a second tube coupling hole **123** connected to the first tube coupling hole **113** of the reset tube **110**, a rotation prevention hole **124** formed as a free end along the circumferential direction on one side of the ratchet tube **120** so that it is coupled to a tube stop protrusion **173** formed on the inner circumferential surface of the anchor **170**, and a ratchet tube connection hole **125** protruding at one end of the reset tube **110**.

(25) The scale drum **130** includes a drum guide groove **131** formed along the lengthwise direction on the inner surface to match the drum guide protrusion formed in the ratchet tube **120**, and a drum rotation groove **132** formed along a spiral shape on the outer circumferential surface to match a drum rotation protrusion **11** protruding in a spiral shape on the inner surface of the housing **10**.

(26) The torsion spring **140** has two ends coupled to the spring base **150** and the ratchet tube **120**, respectively, and during the rotation through the dose dial **20**, one side **141** of the torsion spring **140** fixed to the ratchet tube **120** rotates, while the other side **142** of the torsion spring **140** fixed to the spring base **150** is kept in the fixed state, to get the spring twisted.

(27) The button plate **160** is in the shape of a circular plate fixed and positioned on a first chamber formed on one side with respect to a dial partition wall of the dose dial **20**, and includes a guide fixing hole **161** formed at the center of the button plate **160** and a plurality of button guide holes **162** radially formed with respect to the guide fixing hole. The plurality of button legs **41** that constitutes the injection button **40** is coupled through the plurality of button guide holes. In other words, the button plate **160** allows a stable linear motion of the injection button **40** and stable

placement of the button spring **50** on the injection button **40**. The button spring **50** is placed on the button plate **160** and when the injection button **40** is pressed, the button spring **50** is compressed to the lowest point being the button plate **160**, and when the pressure is removed, the button spring **50** restores its original state.

(28) The anchor **170** includes an anchor circumferential protrusion **171** protruding in a ring shape along the circumferential direction on the inner surface, a housing first coupling protrusion **172** protruding in the shape of a free end along the circumferential direction on the outer surface, a tube rotation prevention protrusion **173** stepped in the shape of a teeth along the axial direction on the inner surface, and a housing second coupling protrusion **174** formed along the axial direction on the outer circumferential surface. The anchor circumferential protrusion **171** is fitted into the anchor coupling groove **122** formed on the ratchet tube **120**, and the housing second coupling protrusion **174** is fitted into an anchor sliding portion **13** formed in the inner surface of the housing **10**.

Meanwhile, the tube rotation prevention protrusion **173** is coupled to the rotation prevention hole **124** that constitutes the ratchet tube **120**, and allows the ratchet tube **120** rotatably coupled to the anchor **170** to rotate in one direction only and disables rotation in reverse direction.

(29) In other words, basically, in the unpressed state of the injection button **40**, the anchor **170** is coupled to the anchor sliding portion **13** formed in the housing **10** through the housing first coupling protrusion **172** and the housing second coupling protrusion **174** to prevent its rotation. In this state, even in case that the reset tube **110** and the ratchet tube **120** are rotated by the operation of the dose dial, the anchor is in a fixed state without rotation.

(30) Meanwhile, when the drug dose setting by the operation of the dose dial is completed, it is possible to prevent the reverse rotation of the ratchet tube **120** through the coupling between the tube rotation prevention protrusion **173** and the rotation prevention hole **124** of the ratchet tube **120**, thereby temporarily preventing the untwisting of the twisted torsion spring.

(31) Referring to FIGS. **3** to **9**, the anchor **170** is rotatably coupled to the anchor coupling groove **122** formed on the ratchet tube **120** through the anchor circumferential protrusion **171**. Meanwhile, the anchor **170** is decouplably coupled to the anchor sliding portion **13** formed in the housing **10** through the housing first coupling protrusion **172** and the housing second coupling protrusion **174**.

(32) In other words, the anchor **170** is configured to slide up and down with the ratchet tube **120** when the injection button **40** is pressed, and the anchor **170** coupled to the ratchet tube **120** provides a structure that allows only the ratchet tube **120** to independently rotate in an immovable state when engaged with the internal groove of the housing **10**, thereby preventing the untwisting of the torsion spring.

(33) In other words, the anchor performs the function of temporarily maintaining the twist of the torsion spring coupled to the ratchet tube **120**.

(34) Referring to FIGS. **10** to **12**, the spring base **150** is a structure which is positioned inside of the dose dial **20** and fixedly coupled to the housing **10** without movement, and includes a base spring stop groove **151** that allows coupling with the end of the torsion spring **140**, and a ratchet **152** configured to generate a clicking sound when engaging with ratchet protrusions **22** formed in the dose dial **20** during the rotation of the dose dial **20**. In other words, feedback on the injection amount is provided through the sequential contact between the ratchet and the ratchet protrusions **22** formed on one surface of the dial partition wall **21** formed along the radial direction in the dose dial **20** by rotating the dose dial **20**.

(35) Referring to FIGS. **14** to **16**, the button plate **160** is positioned on the first dial chamber **20a** formed on one side with respect to the dial partition wall **21** formed in the dose dial **20**, and the injection button **40** is slidably coupled through the plurality of button guide holes **162** formed in the button plate **160**. The plurality of button guide holes **162** may be radially arranged at a predetermined distance with respect to the guide fixing hole **161**. The plurality of button legs **41** has leg stop ends of a curved structure at the end to prevent the slip from the button guide holes **162**.

(36) The button plate **160** is fixed to a plate fixing step **24** formed along the circumferential direction on the inner surface of the dose dial **20** on the first dial chamber **20a**.

(37) In other words, the button plate **160** is coupled moveably with the rotation of the dose dial **20**. In this state, the injection button **40** rotates simultaneously with the rotation of the dose dial **20**, but allows a stable pushing motion through the button plate **160**.

(38) In the unpressed state of the injection button **40**, the dose dial **20** is rotated as follows.

(39) The reset tube **110** coupled to a tube coupling teeth formed along the circumferential direction on the inner surface of the dose dial **20** is coupled to the ratchet tube **120** coupled through the other side of the reset tube **110** and is coupled to the scale drum **130** slidably coupled along a drum guide rail positioned in a straight line shape along the axial direction on the outer surface of the ratchet tube **120**, and the scale drum **130** is coupled to match a drum guide spiral protrusion spirally formed on the inner surface of the housing **10**. Through this, as a consequence, the rotation of the dose dial **20** allows the scale drum **130** to move a predetermined distance along the housing **10** and storage of the twisting moment of the torsion spring **140**.

(40) The drug delivery module **200** includes the rod guide **210** switchably coupled to the reset tube **110** by the operation of the injection button **40**, the piston rod **220** screw-coupled to the outer surface of the rod guide **210** to deliver the drug from the cartridge embedded in the cartridge holder **30** by making a linear motion with the rotational motion of the rod guide **210**, a rod holder **230** coupled to the cartridge holder **30** and through which the piston rod **220** is coupled along the axial direction, and a rod lock **240** screw-coupled to the outer surface of the piston rod **220** in the rod holder **230**.

(41) The rod guide **210** includes a guide body **211** having a guide spiral protrusion **212** separably coupled to a guide coupling groove formed in the inner surface of the piston rod **220**, a guide rotation protrusion **213** protruding on the outer surface of one side of the guide body and separably coupled to a guide coupling groove formed on the inner surface of the reset tube **110**, and a guide ball **214** protruding along the axial direction at the end of the guide body and coupled to a guide fixing hole formed at the center of the button plate **160**. In other words, when the rod guide **210** is in a rotatable state in place without vertical movement by the guide ball coupled through the guide fixing hole of the button plate **160** fixedly coupled to the dose dial **20**, the rod guide **210** is switchably coupled to or decoupled from the reset tube **110** through the guide rotation protrusion, and allows a relative motion by screw coupling with the piston rod **220** through the guide spiral protrusion.

(42) The piston rod **220** has a hollow cylindrical shape through which the rod guide **210** is coupled along the axial direction, and includes a rod movement portion **221** having a lock coupling protrusion to which the rod lock **240** is rotatably coupled and a rod stop portion **222** having a stop step **223** to stop the rod lock **240** from rotating.

(43) The rod holder **230** has a hollow cylindrical shape, and is fixed to the outer side of the cartridge holder **30** through a holder fixing step **231** protruding along the circumferential direction on the outer circumferential surface, and allows a linear motion of the rod lock **240** through a lock guide **232** formed therethrough along the lengthwise direction on the outer circumferential surface.

(44) The rod lock **240** is screw-coupled to the outer surface of the piston rod **220** to allow a rotational motion with the linear motion of the piston rod **220**.

(45) The rod lock **240** includes a lock body **241** coupled to the outer surface of the piston rod **220**, and a lock protrusion **243** protruding along the axial direction on one side of the lock body. The lock protrusion makes a linear motion along the axial direction of the cartridge holder **30** when it is inserted into a lock guide **232**. According to the above-described structure, the rod lock **240** moves with the rotation of the piston rod **220**, but only can make a linear motion along the lock guide **232** under restricted rotation by the lock guide **232**.

(46) In the present disclosure, the rod guide is screw-coupled to the inside of the piston rod having the internal screw, and when the rod guide coupled to the reset tube is rotated in place with the

rotation of the reset tube, the piston rod engaged with the rod guide is moved down to the cartridge holder.

(47) The rod lock is rotatably connected to the external screw of the piston rod, and in this state, the rod holder is installed around the piston rod and the rod lock to allow the piston rod to rotate and the rod lock to make a linear motion.

(48) When the injection button **40** operates, the rod guide **210** coupled to the reset tube **110** is connected to the inside of the piston rod **220** by screw rotation coupling to allow a sub-linear motion of the piston rod **220** through the rotation of the rod guide **210**.

(49) The drug injection control device according to the present disclosure as described above allows a structure in which the rod guide **210** rotates and the piston rod moving with the rod guide **210** makes a linear motion to stably inject the drug set through the dose dial.

(50) The foregoing description is made to describe the technical spirit of the present disclosure for illustrative purposes, and it is obvious to those skilled in the art that a variety of modifications and change may be made thereto without departing from the essential features of the present disclosure. Therefore, the embodiments disclosed herein are provided to describe the technical spirit of the present disclosure and not intended to be limiting, and the technical spirit and scope of the present disclosure is not limited by the embodiments. The scope of protection of the present disclosure should be interpreted by the appended claims, and it should be interpreted that the scope of protection of the present disclosure covers all the technical spirit within the equivalent scope.

Claims

1. A drug injection control device comprising: a housing being hollow; a dose dial rotatably coupled to a side of the housing; a cartridge holder fixedly coupled to an opposite side of the housing, wherein a cartridge containing a drug is configured to be received in the cartridge holder, and an injection needle is configured to be coupled to a side to deliver the drug; an injection button positioned in the dose dial; an injection amount setting module to set an amount of the drug to be injected by storing a twisting moment in a torsion spring positioned in the housing; and a drug delivery module to deliver the drug from the cartridge embedded in the cartridge holder by moving through the release of the twisting moment applied to the torsion spring, wherein the injection amount setting module includes a reset tube switchably coupled to the dose dial by the operation of the injection button, a ratchet tube coupled to an outer side of the reset tube to rotate at the same time, a scale drum which slidably rotates along an axial direction on an outer surface of the ratchet tube, the torsion spring having two ends coupled to the housing and the ratchet tube, and an anchor rotatably coupled to the ratchet tube and slidably coupled to an inside of the housing, wherein the anchor includes an anchor circumferential protrusion protruding along a circumferential direction on an inner surface, a tube rotation prevention protrusion stepped in a teeth shape along the axial direction on the inner surface, and housing coupling protrusions formed along the axial direction on the outer circumferential surface, and wherein the tube rotation prevention protrusion is coupled to a rotation prevention hole which constitutes the ratchet tube to allow the ratchet tube rotatably coupled to the anchor to rotate in one direction only.
2. The drug injection control device according to claim 1, wherein the anchor is rotatably coupled to an anchor coupling groove formed on the ratchet tube through the anchor circumferential protrusion, and is decouplably coupled to an anchor sliding portion formed in the housing through the housing coupling protrusions.
3. The drug injection control device according to claim 1, wherein a rod guide coupled to the reset tube is connected to an inside of a piston rod by screw rotation coupling to allow a sub-linear motion of the piston rod through the rotation of the rod guide, thereby having a cartridge drug release structure through the screw coupling between the piston rod and the rod guide.
4. The drug injection control device according to claim 1, wherein the torsion spring has the two

ends coupled to a spring base immovably coupled to the housing and the ratchet tube, respectively, and wherein during the rotation through the dose dial, a side of the torsion spring fixed to the ratchet tube rotates, while an opposite side of the torsion spring fixed to the spring base is kept in fixed state, to get the spring twisted.

5. The drug injection control device according to claim 1, wherein a button plate is fixed to a plate fixing step formed along the circumferential direction on an inner surface of the dose dial on a first dial chamber formed through a partition wall in the dose dial.

6. The drug injection control device according to claim 1, wherein the injection amount setting module includes a button plate fixed and positioned in the dose dial to perform a function of supporting a button spring, wherein the injection button is slidably coupled through the button plate, wherein the drug delivery module includes: a rod guide switchably coupled to the injection amount setting module by the operation of the injection button, and a piston rod screw-coupled to an outer side of the rod guide to deliver the drug from the cartridge embedded in the cartridge holder by making a linear motion with a rotational motion of the rod guide, and wherein the injection button rotates simultaneously with the rotation of the dose dial, but allows a stable pushing motion through the button plate.
